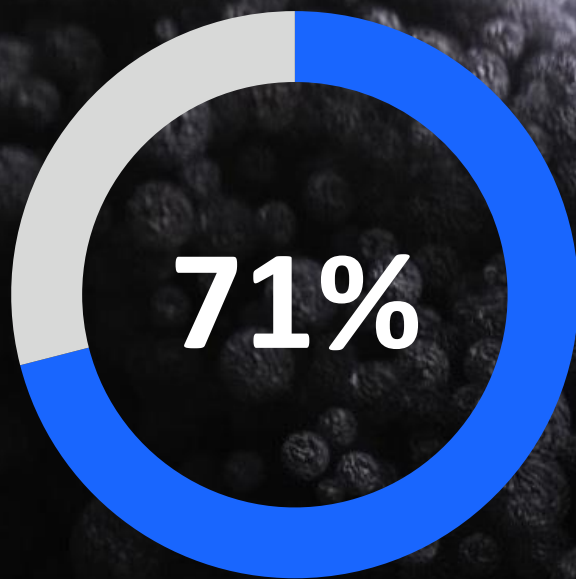


WORKFLOW ESSENTIALS: AUTOMATING TO SOLVE CUSTOMER USE CASES

DYNATRACE SOLUTIONS ENGINEER
TRAINING

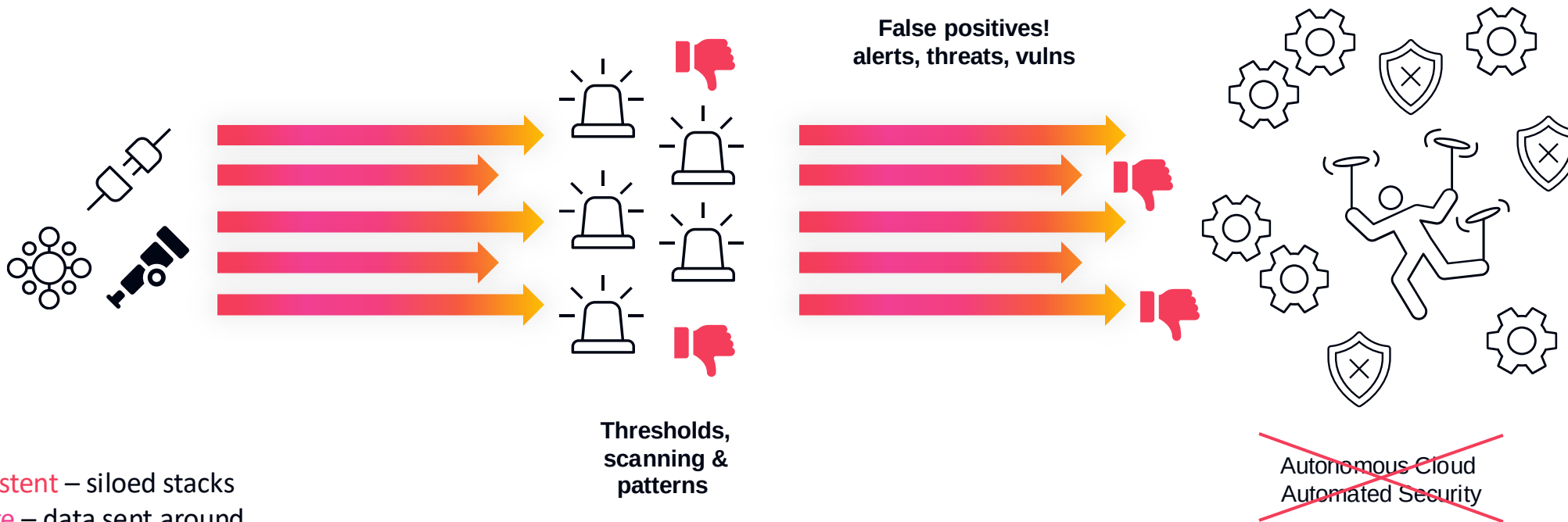
AGENDA

- Goal: Have the foundation to be able to automate any customer use case within Dynatrace using Workflow automations
- The value of automation
- Customer use case examples
- Workflow Essentials
 - Parameters and Task Results
 - Fetch API for HTTP
 - Looping Tasks
 - Leveraging the Dynatrace SDK
 - Third-party Integrations
- Best practices
- Assessment Action Items



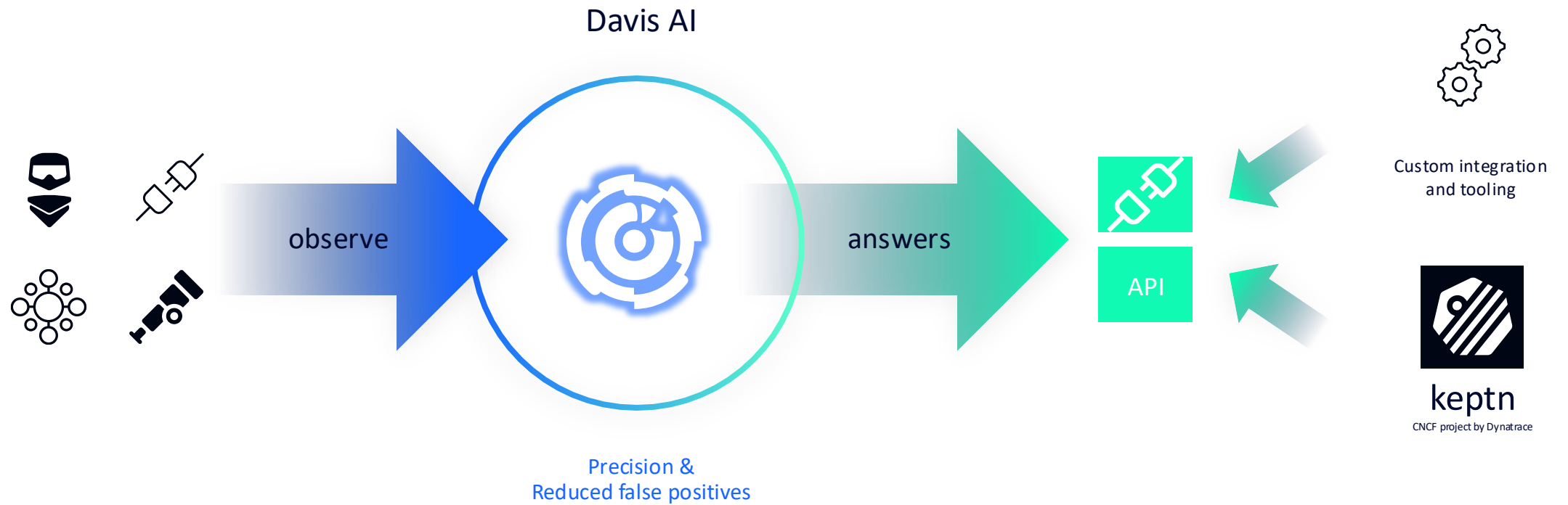
of **CIOs** say the **explosion of data** produced by cloud-native technology stacks is **beyond human ability to manage**.

AUTOMATION IS CHALLENGING

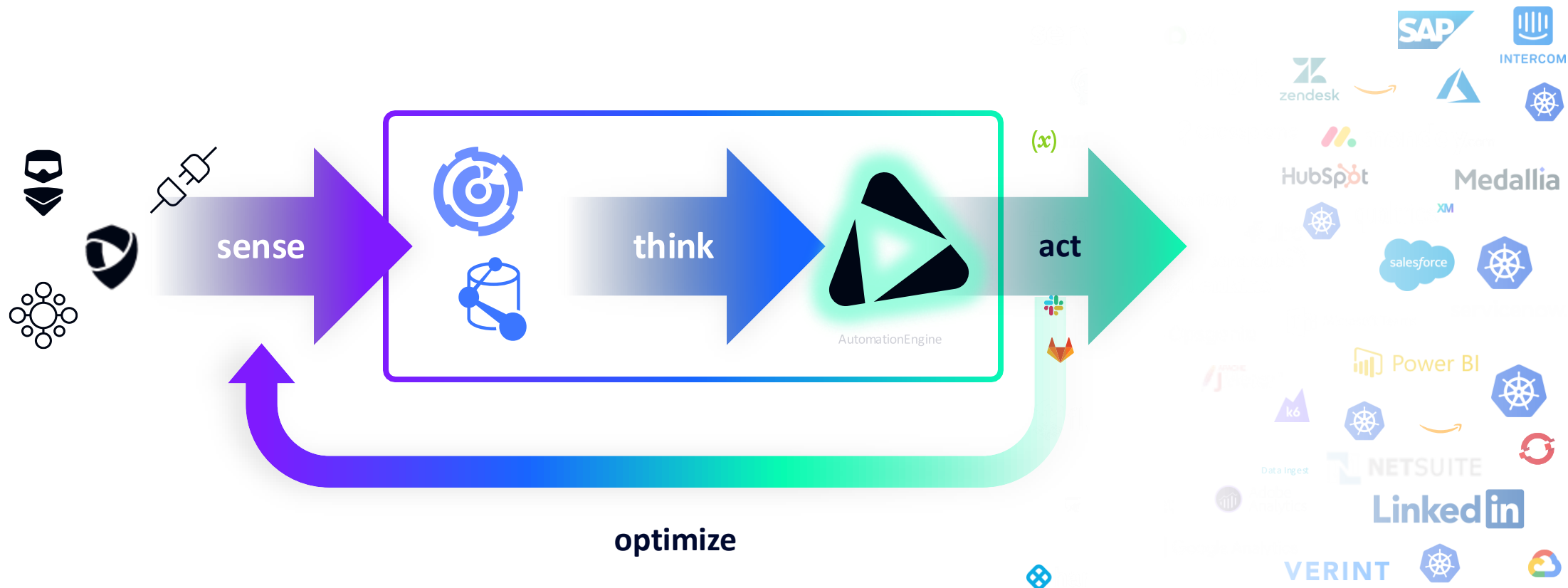


- Inconsistent** – siloed stacks
- Insecure** – data sent around
- Inefficient** – plumbing integrations
- Impossible** – on false positives
- Intricate** – 1000+ CNCF tools

AUTOMATIC ANSWERS FROM DYNATRACE



ANSWER-DRIVEN AUTOMATION



AUTOMATIONENGINE™



LAUNCHED in April 2023

Intelligent automation for observability and security at scale

Prevent | Operate | Remediate | Protect | Accelerate

Visual Workflows

Intuitive ad-hoc automation, drag-n-drop flow-logic, schedules, orchestration, workflow execution monitoring and control

Cloud-Native as-Code Workflows

Declarative GitOps automation-as-code, for Kubernetes and Hyperscalers

Answer-driven Automation

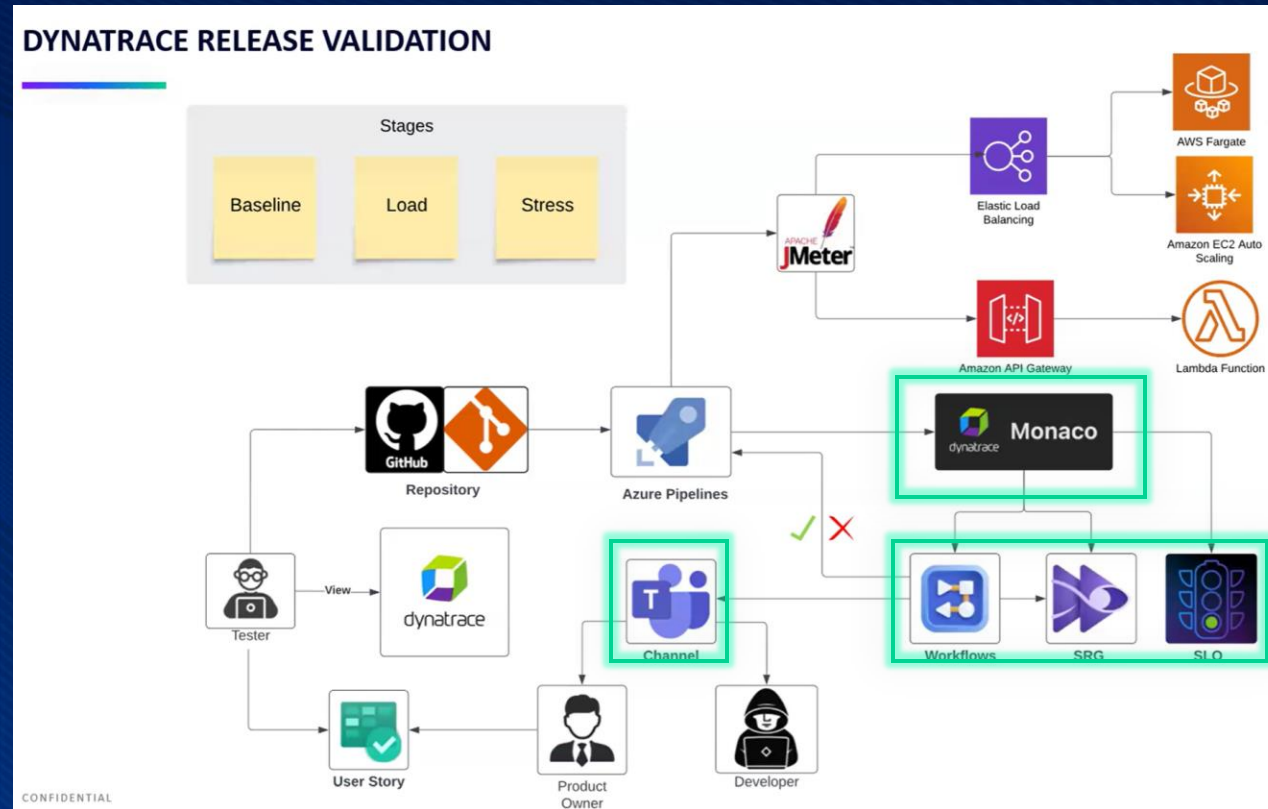
Intelligent automation and orchestration based on observability and security analytics.

Automated Financial Services Release Validation in Performance Testing at NEQUI

FinTech in Colombia, started in 2016 with 300 customers up to 18.6M in 2023.

Problem statement:

- Product teams create user stories w/o SLOs
- Developers don't know about performance impact on infrastructure
- Testers are flying blind (e.g., no e2e view, no traces, latency and no correlation)
- Difficulty in communicating test analysis
- Just JMeter is automated



What has been achieved:

- SLO-centric view by product teams
- Developers can align development efforts towards infra and financial goals.
- Testers have a shared viewpoint, make data-driven decisions, and get understanding of services behavior
- Product teams get immediate feedback on SLO breaches
- Automation made performance testing **30% faster**

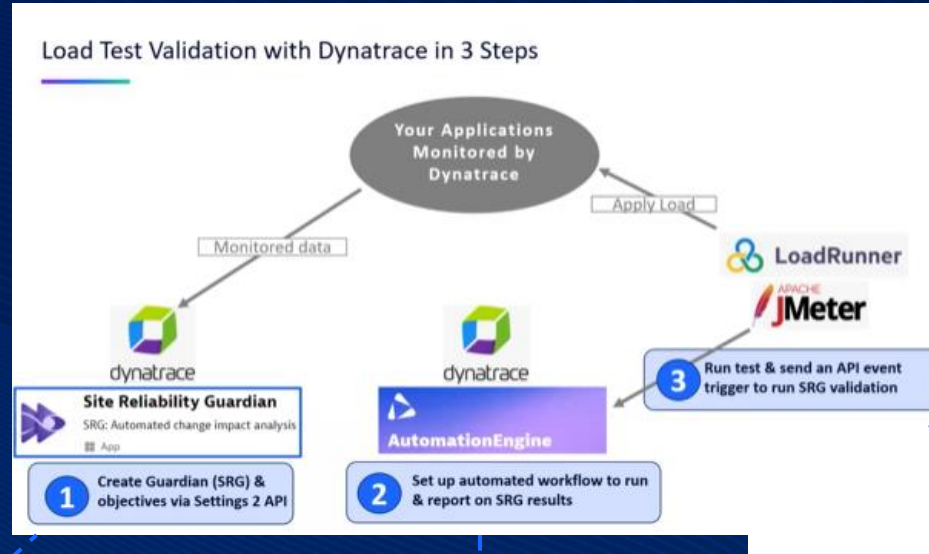
Automating Performance Testing at Scale! Democratize Observability at



Leader in the US Mortgage Market, founded in 1986, 7,000 employees.

Problems:

- Stale regression & load tests for 40+ applications
- Centralized & small performance testing team
- Manual test result reporting process
- Skepticism & confusion about test results
- No root cause diagnostics



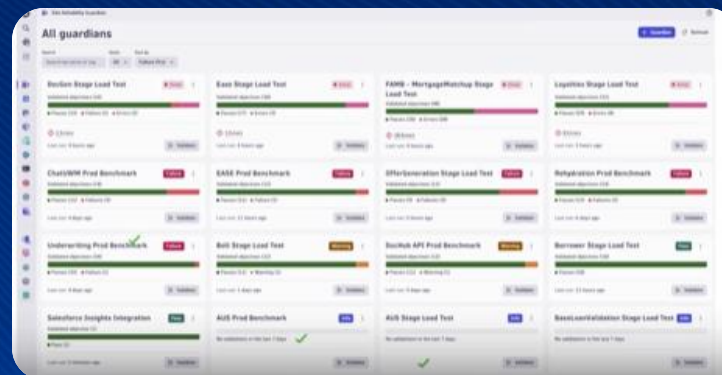
What has been achieved:

- Quickly implement a large-scale performance test strategy
- Decentralize our performance test team & accelerate product team autonomy
- Improve application observability in Dynatrace

(1) Config-as-code to enable scalability

```
JSON Content
{
  "insertAfter": null,
  "insertBefore": null,
  "schemaVersion": "1.1",
  "updateToken": null,
  "value": {
    "name": "Load Documents Stage Load Test",
    "tags": {},
    "variables": {},
    "objectives": [
      {
        "name": "dql test",
        "objectiveType": "DQL",
        "dqlQuery": "Fetch events\\n| filter matchesValue(event.type, \\\"PROCESS_RESTART\\\")\\n| filter matchesPhrase (R.entity._process_group_instance.name, \\\"SERVICE-6301D640BFE72\\\")\\n| summarize total = count()",
        "comparisonOperator": "GREATER_THAN_OR_EQUAL"
      }
    ]
  }
}
```

(2) Site Reliability Guardians for 70+ App teams



(3) Load test pattern



Automating Dynatrace tagging, ownership, and alerting for Kubernetes at



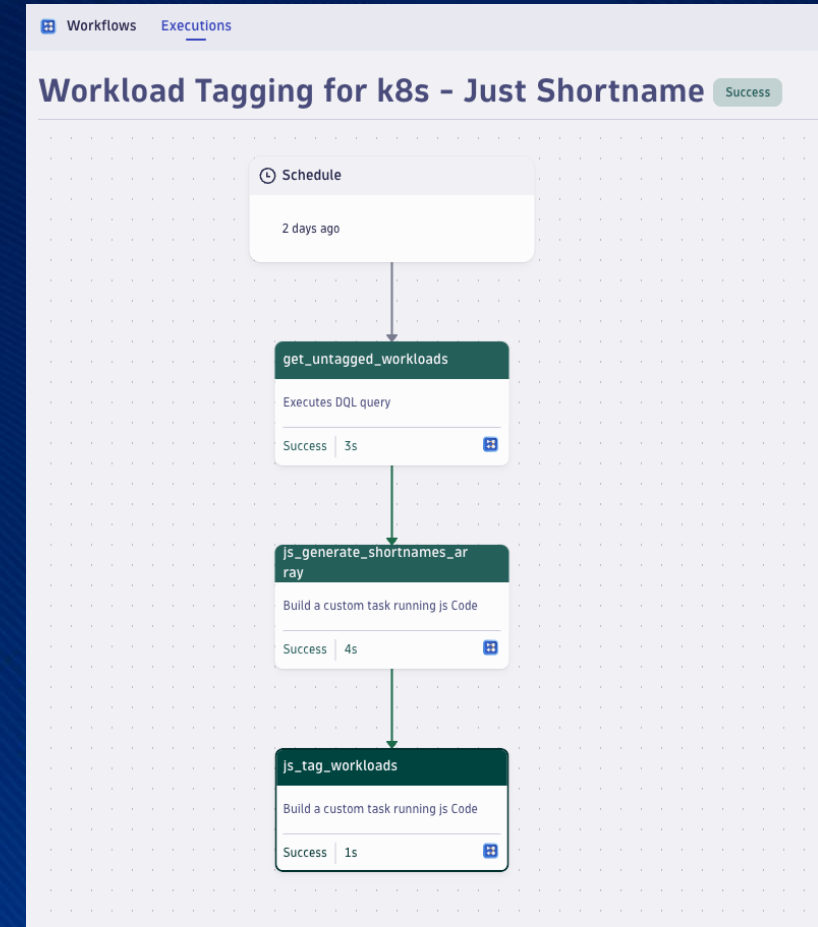
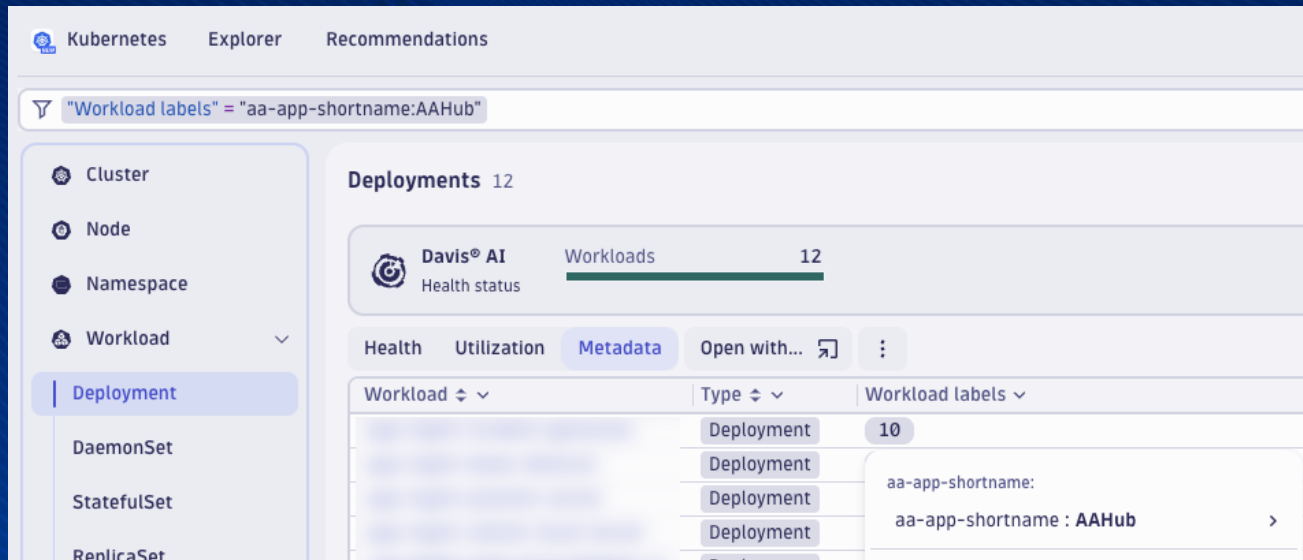
Major US airline based in Texas. It is the largest airline in the world when measured by scheduled passengers carried, revenue passenger mile, and daily flights

Problems:

- Dynatrace drives alert notifications through tagging and ownership mechanisms
- Customer defines ownership for Kubernetes based on their own label
- Manual tag propagation is a high level of effort and results in stale config

What has been achieved:

- Entities missing required tag data are queried through DQL
- Tags are applied to every workload dynamically
- Alert notifications are automatically handled correctly through tagging that is applied and updated every hour; saving AA many people-hours per week



CONFIDENTIAL

**TIME TO GET
HANDS ON**

Overview

Materials

Environments

Attendees

Lab environments

Validate that you can access your Dynatrace environment through the Dynatrace University Event

Dynatrace Environment

You must accept the invitation to the Dynatrace account to proceed. Please check your inbox or spam folder for the emailed invitation.

Username

[View environment](#)

[Trouble logging in?](#) 

Note: your event page may look different than the screenshot above

ENVIRONMENT SETUP

Prerequisites

Click on Settings and choose
Authorization settings

Authorization

To run a workflow with your user as the actor (this is the default on workflow creation), you need to authorize the AutomationEngine to use permissions in your name. Note: The AutomationEngine authorization only works for permissions granted to your user.

Discard

Save

Tony Pope-cruz

☒ Workflow admin

Authorization settings

Scheduling

Rules

Calendars

Enable all primary
permissions

Primary permissions

It's recommended to grant at least the following permission to be able to run workflows.

- ☒ app-engine:apps:run - Grants permission to list and run apps and gives basic access to the Launcher
- ☒ app-engine:functions:run - Grants permission to use the function-executor
- ☒ app-settings:objects:read - Grants permission to read app settings objects belonging to the schema

Secondary permissions

 [35 warnings](#)

You can authorize the permissions below to expand or limit your workflows.

Search

search for a permission...

Select all secondary
permissions

☒ Select all

> ☒  app-engine

Locate and open the Hub App

Hub

Apps (1)

- Hub

Settings (2)

- Hub subscriptions
- Installation Requests

Hub (8)

- Hub
- Azure IoT Hub
- Azure Notification Hub Namespaces

Not what you're looking for?

- Search in a different category
- Configure search results
- Search support resources
- Share your feedback

Help and tips

Search all apps, extensions, and technologies

Here you'll find hundreds of technologies to tame digital complexity and get more value from data, from application and infrastructure monitoring to digital experience and application security.

Search and locate the App 'Slack for Workflows'

Hub home

Get data into Dynatrace

Build your own app

All (6)

Log Management and Analytics

Infrastructure Observability

Automations

Application Observability

Application Security

Digital Experience

Business Analytics

Slack for Workflows

App

Actions

↺



Slack for Workflows

App 1 Actions

Automate Slack messaging for security incidents, attacks, remediation, and more.



Microsoft 365 for Workflows (Preview)

App 1 Actions

Send targeted emails from workflows using your Microsoft 365 account.



Davis® for Workflows

App 1 Actions

Implement custom AI analysis in Grail data lakehouse using Davis® for Workflows.



Email for Workflows (Preview)

App 1 Actions

Send out-of-the-box targeted Emails with workflows.



Ownership Installed

App 3 Actions

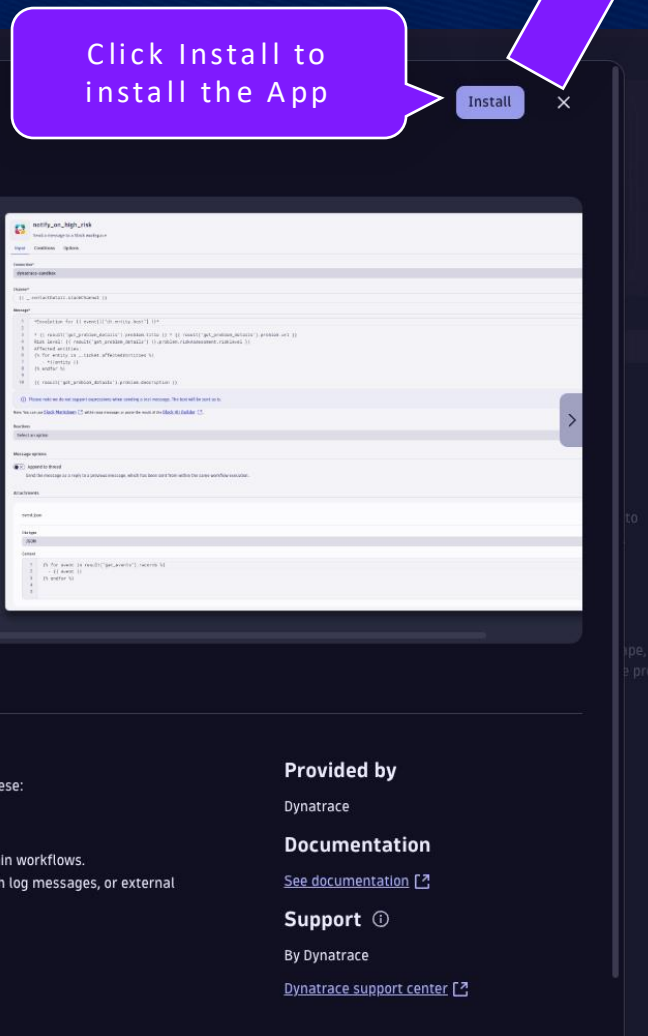
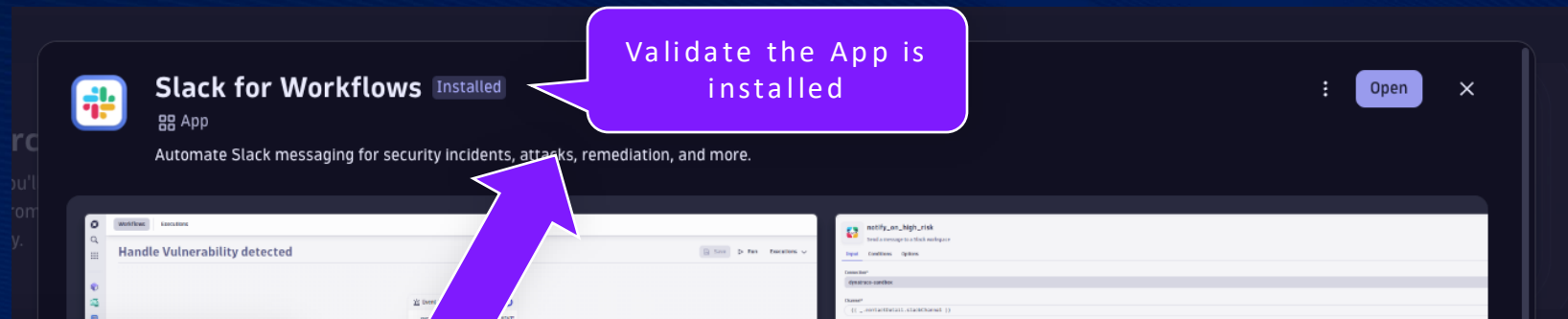
Enables the import of and access to ownership information of entities.



Workflows Installed

App 3 Actions

Automate tasks in your IT landscape, remediate problems, and visualize processes



Import the workflow 'Workflow Essentials - Config Generator'

Import workflow "Workflow Essentials - Config Generator" ×

Required apps

For this workflow to be fully functional, you need to have the required apps installed and the proper connections configured.



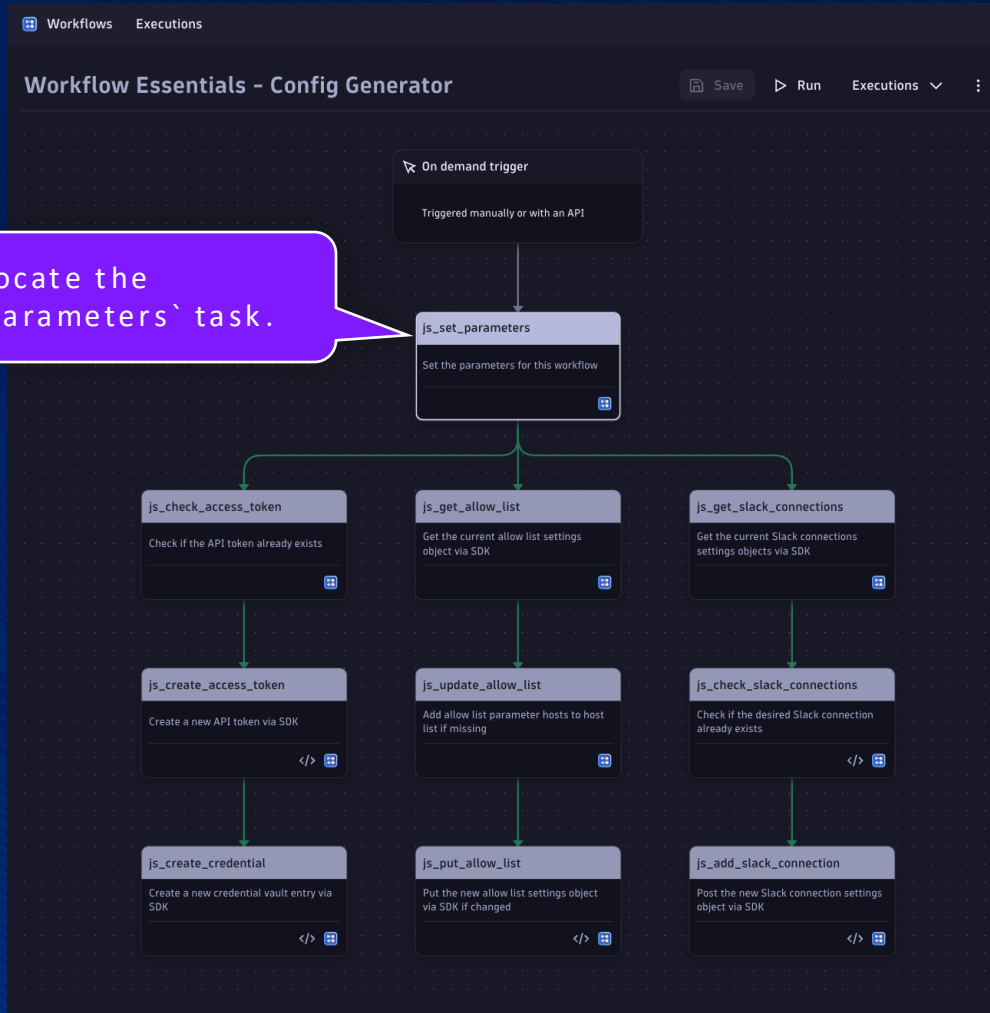
Workflows 1.840.0 installed

Process automation for Dev, Ops, or even DevOps.

 Refresh

Cancel

Import



Locate the
`js\_set\_parameters` task.

js_set_parameters

Set the parameters for this workflow

Input Conditions Options Sample result

Source code

```
1 import { CredentialsScope, CredentialsScopesItem, CredentialsType } from "@dynatrace-sdk/client"
2
3 export default async function () {
4
5   let PARAMETERS = {
6
7     CREDENTIAL: {
8
9       // TOKEN_NAME - the name of the Dynatrace API access token
10      TOKEN_NAME: "workflow-essentials-metric-ingest", // i.e. TOKEN_NAME: "workflow-essentials
11
12      // TOKEN_SCOPES - an array of double quoted string token scopes
13      TOKEN_SCOPES: ["metrics.read", "metrics.write", "metrics.ingest"], // i.e. TOKEN_SCOPES: ["
14
15      // CREDENTIAL_NAME - the name of the Dynatrace credential vault entry
16      CREDENTIAL_NAME: "workflow-essentials-metric-ingest", // i.e. CREDENTIAL_NAME: "workflow-
17
18      // CREDENTIAL_SCOPES
19      CREDENTIAL_SCOPES: [CredentialsScopesItem.AppEngine],
20
21      // CREDENTIAL_ACCESS
22      CREDENTIAL_ACCESS: false
23    },
24
25    ALLOW_LIST: {
26
27      // ALLOW - a list of hostnames to allow connections from
28      ALLOW: ["api"],
29
30    },
31
32    SLACK: {
33
34      // CONNECTION NAME - the name of the Slack Connection
35      CONNECTION_NAME: "workflow_essentials_demo",
36
37      // CONNECTION_TOKEN - the Bot User OAuth Token of the Slack App
38      CONNECTION_TOKEN: ""
39    }
40  }
41 }
```

Locate the code snippet for
the Slack connection.

```
SLACK: {

  // CONNECTION NAME - the name of the Slack Connection
  CONNECTION_NAME: "workflow_essentials_demo",

  // CONNECTION_TOKEN - the Bot User OAuth Token of the Slack App
  CONNECTION_TOKEN: "xoxb-
```

Set the
`CONNECTION\_TOKEN`
value to the Slack App
Bot User Token

Workflow Essentials – Config Gener...

Success

Edit workflow

Cancel

Refresh



Run the workflow and validate that it runs successfully.

js_check_access_token

Check if the API token already exists

Success 5s

js_create_access_token

Create a new API token via SDK

Success 1s

js_create_credential

Create a new credential vault entry via SDK

Success 1s

js_get_allow_list

Get the current allow list settings object via SDK

Success 5s

js_update_allow_list

Add allow list parameter hosts to host list if missing

Success 1s

js_put_allow_list

Put the new allow list settings object via SDK if changed

Success 1s

js_get_slack_connections

Get the current Slack connections settings objects via SDK

Success 2s

js_check_slack_connections

Check if the desired Slack connection already exists

Success 2s

js_add_slack_connection

Post the new Slack connection settings object via SDK

Success 4s



js_create_credential

Create a new credential vault entry via SDK

Detail Input **Result**

credential

1

CREDENTIALS_VAULT-4D84636186383332



Copy the credential vault ID from the result of the `js\_create\_credential` task

Slack Connections

Authentication details for Slack API

[More...](#)

Add item

A Slack Connection has been created

Summary

workflow_essentials_demo

Connection name

workflow_essentials_demo

The name of the Slack connection

Bot token

Change

The bot token obtained from the Slack App Management UI

Limit outbound connections

You can limit the accessibility of public endpoints from functions running in the Dynatrace JavaScript runtime, for example, the backends of apps and functions written in the Dashboards, Notebooks and Automations app.

☒ Limit outbound connections to endpoints in the allow-list

Allow-list

Add item

The allow list for outbound connections has been updated

Filter items...

Allowed Host

*.amazonaws.com

*.sprint.dynatracelabs.com

api.status.io

files.slack.com

slack.com

webhook.site

Import the workflow
'Workflow Essentials - Data
Generator'

Import workflow "Workflow Essentials - Data Generator"

×

Required apps

For this workflow to be fully functional, you need to have the required apps installed and the proper connections configured.



Workflows

1.840.0

installed

Process automation for Dev, Ops, or even DevOps.

↻ Refresh

Cancel

Import

Workflow Essentials - Data Generator

Save Run Executions

No changes are required. The workflow is scheduled to run every 5 minutes.

Schedule

Every 5 mins | Europe/London

config

Build a custom task running js Code

check_if_data_exists

Executes DQL query

generate_historical_data

Build a custom task running js Code

generate_new_data

Build a custom task running js Code

process_temporary_events

Build a custom task running js Code

Time interval trigger

Run workflow at fixed time intervals.

Change trigger

Run every (mins)*

5

Minimum 1 minute, maximum 720 minutes

Rule

Every day

[Create a new rule](#)

Advanced options

Next run: 9/3/2024, 10:25:00 AM -05:00

Show next 10 runs

WORKFLOW ESSENTIALS

Workflow Exercise 1

WORKFLOW ESSENTIALS

Goal: Utilize essential Workflow techniques applicable to most use cases

- Parameters and Task Results
 - Parameter Task – define parameters/variables that will be used by subsequent tasks
 - Parameter Task Output – return parameters for subsequent tasks
 - Accessing Parameter Task Results – access the parameters from result of the task
- Fetch API
 - HTTP GET (Dynatrace Status.io) – bring external data into Dynatrace workflows
 - HTTP POST (Webhook.Site) – send data to external systems from Dynatrace workflows
- Looping Tasks
 - DQL Query – execute a DQL query that returns multiple records/rows
 - Looping Over Results – perform logic against each DQL record/row

Workflow Essentials

Modified

Save

Run

Executions ▾



Begin by creating a new Workflow in the Workflows App within your Dynatrace environment.

<⚡> Trigger

Click to select a trigger



Select a trigger: choose 'On demand trigger'

Select a trigger

Triggers start the execution of a workflow. They can listen for events or be scheduled to run at fixed times or intervals.

Event



Event trigger

Run workflow based on a custom event filter.



Davis problem trigger

Run workflow based on problems detected by Davis.



Davis event trigger

Run workflow based on events detected by Davis.

Schedule



Fixed time trigger

Run workflow at a fixed time of day.



Time interval trigger

Run workflow at fixed time intervals.



Cron schedule trigger

Run workflow on a cron schedule.

On demand



On demand trigger

Use the Run button to trigger the workflow or use an API.

Workflow Essentials

Modified

Save

Run

Executions ▾

⋮

On demand trigger

Triggered manually or with an API

+

Click the '+' button to
add a new action to
the Workflow.

On demand trigger

↩ Change trigger

⋮

Select Run to manually trigger the workflow or use an [API](#). [↗](#)

API script

```
1 // This is an example of a REST request using curl
2 curl \
3   -H 'Authorization: Bearer <YOUR_BEARER_TOKEN>' \
4   -X POST https://hfn13693.apps.dynatrace.com/platform/
```

Note: Use your [OAuth2 client configuration](#) [↗](#) to get a valid bearer token for the request. It will require the `automation:workflows:run` scope.

Workflow Essentials

Modified

Save

Run

Executions



On demand trigger

Triggered manually or with an API

New task

Select an action from the list



Choose action: choose
`Run JavaScript` action
type.

Choose action

Add a task to the workflow and select the action it performs

Search actions...

Workflows



Execute DQL Query

Executes DQL query



HTTP Request

Issue an HTTP request to any API



Run JavaScript

Build a custom task running js Code

Kubernetes Automations (Preview)



Apply configurations

Apply a configuration change to a resource or create new Kubernetes resources



Delete resources

Delete Kubernetes resources



Get logs

Get the logs for a container in a pod (limited to max. 1MB)



Get resource

Retrieve a Kubernetes resource



List resources

List Kubernetes resources



Restart deployment

Restart a Kubernetes deployment

Workflow Essentials

Save

Run

Executions ▾



Paste the following code snippet
into the action `Input`:

On demand trigger

Triggered manually or with an API

js_set_paramet

Set the paramete

Set the value of the
missing variables



js_set_parameters

Set the parameters for this workflow

Change action



Input

Conditions

Options

Sample result

Source code

```
1 export default async function () {
2
3   let PARAMETERS = {
4
5     // INITIALS - the initials of the user for this lab
6     INITIALS: "TPC", // i.e. INITIALS: "TPC"; the initials of the user for this la
7
8     // LOCATION - the city and country of the user for this lab
9     LOCATION: "Dallas, United States", // i.e. LOCATION: "Dallas, United States";
10
11    // TENURE - the number of years of tenure of the user for this lab
12    TENURE: 14, // i.e. TENURE: 14; the user for this lab has 14 years of tenure
13
14    // STATUS_URL - the URL of the dynatrace status.io API
15    STATUS_URL: "https://api.status.io/1.0/status/546d8cb6af8407b6730000cb", // i
16
17    // WEBHOOK_URL- the URL of the webhook.site unique endpoint
18    WEBHOOK_URL: "https://webhook.site/18637ec8-fcf9-43ec-86bf-993626982c6c", //
19
20  }
21
22  return PARAMETERS;
23 }
```

Workflow Essentials

Save

Run

Executions



On demand trigger

Triggered manually or with an API

js_set_parameters

Set the parameters for this workflow



Choose action: choose
`Run JavaScript` action
type.

js_output_parameters

Output the parameters from the
previous task

Click the `+` button to
add a new action to
the Workflow.



js_output_parameters

Output the parameters from the previous task

Change action



Input

Conditions

Options

Sample result

Paste the following code
snippet into the action
`Input`:

Source code

```
1 import { execution } from '@dynatrace-sdk/automation-utils';
2
3 const PARAMETERS_TASK = 'js_set_parameters';
4
5 export default async function ({ execution_id }) {
6
7     // get parameters from previous tasks
```



js_output_parameters

Output the parameters from the previous task

Change action



Input

Conditions

Options

Sample result

Run this task if

js_set_parameters

Set the task conditions

success



And custom condition was met

1

Else

Stop here



Workflow Essentials

Success

Edit workflow

Cancel

Refresh



On demand trigger

5 seconds ago

js_set_parameters

Set the parameters for this workflow

Success | 1s

js_output_parameters

Output the parameters from the previous task

Success | 1s

Run the workflow and validate the results

js_output_parameters log

```
1 {
2   TENURE: 14,
3   INITIALS: "TPC",
4   LOCATION: "Dallas, United States",
5   STATUS_URL: "https://api.status.io/1.0/status/546d8cb6af8407b6730000cb",
6   WEBHOOK_URL: "https://webhook.site/a356660b-aec3-4eba-87fa-770dd4d7719b"
7 }
8
```



js_output_parameters

Output the parameters from the previous task

Cancel task

Detail

Input

Result

1 The lab user TPC, is located in Dallas, United State

Workflow Essentials

On demand trigger

Triggered manually or with an API

js_set_parameters

Set the parameters for this workflow

js_fetch_status

Read dynatrace status.io using fetch api

Choose action: choose `Run JavaScript` action type.

js_output_parameters

Output the parameters from the previous task

Click the `+` button to add a new action to the Workflow.



js_fetch_status_io

Read dynatrace status.io using fetch api

Change action

Input

Conditions

Options

Sample result

Source code

```
1 import { execution } from '@dynatrace-sdk/automation-utils';
2
3 const PARAMETERS_TASK = 'js_set_parameters';
4
5 export default async function ({ execution_id }) {
6
7   // get parameters from previous tasks
```

Paste the following code snippet into the action `Input`:



js_fetch_status_io

Read dynatrace status.io using fetch api

Change action

Input

Conditions

Options

Sample result

Run this task if

js_set_parameters

Set the task conditions

success



And custom condition was met

1

```
{{ result("js_set_parameters")['STATUS_URL'] is defined }}
```



Else

Stop here

Workflow Essentials

Triggered manually or with an API

js_set_parameters

Set the parameters for this workflow

js_output_parameters

Output the parameters from the previous task

js_fetch_status_io

Read dynatrace status.io using fetch api

js_fetch_webhook_site

Send json content to webhook.site using fetch api

Click the '+' button to add a new action to the Workflow.

Choose action: choose 'Run JavaScript' action type.



js_fetch_webhook_site

Send json content to webhook.site using fetch api

Change action

Input

Conditions

Options

Sample result

Paste the following code snippet into the action 'Input':

Source code

```
1 import { execution } from '@dynatrace-sdk/automation-utils';
2
3 const PARAMETERS_TASK = 'js_set_parameters';
4 const FETCH_STATUS_TASK = 'js_fetch_status_io';
5
6 export default async function ({ execution_id }) {
```



js_fetch_webhook_site

Send json content to webhook.site using fetch api

Change action

Input

Conditions

Options

Sample result

Run this task if

Set the task conditions

js_fetch_status_io

success



And custom condition was met

1

```
{{ result("js_fetch_status_io")['result'] is defined }}
```



Else

Stop here

Workflow Essentials

Success

Edit workflow

Cancel

Refresh



On demand trigger

8 seconds ago

Run the workflow and validate the results

js_set_parameters

Set the parameters for this workflow

Success 1s



js_output_parameters

Output the parameters from the previous task

Success 1s



js_fetch_status_io

Read dynatrace status.io using fetch api

Success 1s



js_fetch_webhook_site

Send json content to webhook.site using fetch api

Success 1s



js_fetch_webhook_site

Send json content to webhook.site using fetch api

Cancel task

Detail Input Result

200

Workflow Essentials

On demand trigger
Triggered manually or with an API

js_set_parameters
Set the parameters for this workflow

js_output_parameters
Output the parameters from the previous task

to using fetch
</>

js_fetch_webhook_site
Send json content to webhook.site using fetch api
</>

dql_query_bizevents
Query bizevents

Choose action: choose
'Execute DQL Query'
action type.

Click the '+' button to
add a new action to
the Workflow.



dql_query_bizevents

Query bizevents

Input Conditions Options Sample result

DQL query

```
1 fetch bizevents, from: now()-2h
2 | filter event.provider == "workflow.essentials.withdraw-processing"
3 | summarize count = count(), by: {step = event.type}
4 | sort step asc
```

To learn more about using DQL, you can visit [Dynatrace Query Language](#)

Fail on empty result

☒ Fail the task when query result is empty

dql_query_bizevents

Query bizevents

Change action

Input Conditions Options Sample result

Run this task if

js_fetch_status_io

Set the task conditions

success

And custom condition was met

1

Else

Stop here

Paste the following DQL
query into the action
'Input':

Workflow Essentials

Save Run Executions

Choose action: choose `Run JavaScript` action type.



Click the `+` button to add a new action to the Workflow.



js_loop_over_results

Execute task for each loop item, bizevent record

Change action

Input Conditions Options Sample result

Source code

```
1 import { execution, actionExecution } from '@dynatrace-sdk/automation-utils';
2 import { getCurrentUserDetails, getEnvironmentId, getEnvironmentUrl } from '@dynatrace-sdk/app-environment';
3
4 const PARAMETERS_TASK = 'js_set_parameters';
5
6 export default async function ({ execution_id, action_execution_id }) {
7
8   // get parameters from previous tasks
9   // execution
10  const ex = await execution(execution_id);
11  // parameters
12  const parameters = await ex.result(PARAMETERS_TASK);
13  // WEBHOOK_URL
14  const WEBHOOK_URL = parameters['WEBHOOK_URL'];
15
16  // get the loop item for the action execution
17  const actionEx = await actionExecution(action_execution_id);
18  const record = actionEx.loopItem['item'];
19  // log the loop item record
20  console.log(JSON.stringify(record));
21  // create variables for the step and count attributes within the record
22  const step = record['step'];
23  const count = record['count'];
24
25  // check if step is "06.funds.received", send webhook with details
26  if(step == "06.funds.received") {
27
28    // get a timestamp to send with the payload
29    // https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Date/Date
30    const timestamp = new Date();
31
32    // Environment ID
33    const environment_id = getEnvironmentId();
34
35    // Workflow execution user details
36    const user_details = getCurrentUserDetails();
37    const user_name = user_details['name']; // get the user name
38
39    // set the fetch() api parameters [method, headers, body]
40    // https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API/Using_Fetch
41    const options = {
42      method: "POST",
43      headers: {
44        'content-type': "application/json",
```

Paste the following code snippet into the action `Input`:



js_loop_over_results

Execute task for each loop item, bizevent record

↩ Change action



Input

Conditions

Options

Sample result

Run this task if

dql_query_bizevents

Set the task conditions

success



And custom condition was met

1

```
{{ result("dql_query_bizevents").records | length > 0 }}
```

Else

Stop here



js_loop_over_results

Execute task for each loop item, bizevent record

↩ Change action



Input

Conditions

Options

Sample result



Wait before



Controls how long a task stays in waiting state before being run. Default set to 0 second.



Loop task



Iterates over items in a list, allowing actions to be executed repeatedly.

Item variable name*

item

Use {{ _item }} to access item in each iteration.

List*

dql_query_bizevents.records



Concurrency

1

By default loops execute sequentially with concurrency set to 1. You can increase how often it runs in parallel.



Retry on error



Automatically rerun the task on failure. Default set to no retries.



Adapt timeout



Maximum time for the task to finish before it is cancelled due to time out. Default set to 15 minutes.

Click on the task's `Options` tab.
Toggle on `Loop task` setting.

Workflow Essentials

Success

Edit workflow

Cancel

Refresh



On demand trigger

2 minutes ago

Run the workflow and validate the results

js_set_parameters

Set the parameters for this workflow

Success 1s

js_output_parameters

Output the parameters from the previous task

Success 1s

js_fetch_status_io

Read dynatrace status.io using fetch api

Success 1s

js_fetch_webhook_site

Send json content to webhook.site using fetch api

Success 2s

dql_query_bizevents

Query bizevents

Success 2s

js_loop_over_results

Execute task for each loop item, bizevent record

Success 9s

js_loop_over_results log

```
28
29 =====
30 {"step": "06.funds.received", "count": "4"}
31
32
33
34 =====
35
```



js_loop_over_results

Execute task for each loop item, bizevent record

Cancel task

Detail Input Result

0	<pre>1 { 2 "step": "01.withdraw.submitted", 3 "count": "83" 4 }</pre>
1	<pre>1 { 2 "step": "02.balance.verified", 3 "count": "76" 4 }</pre>
2	<pre>1 { 2 "step": "03.notify.bank", 3 "count": "73" 4 }</pre>
3	<pre>1 { 2 "step": "04.bank.acknowledge", 3 "count": "94" 4 }</pre>
4	<pre>1 { 2 "step": "05.transfer.funds", 3 "count": "3" 4 }</pre>
5	<pre>1 { 2 "step": "05.transfer.funds.trxn.withdraw-processing.error", 3 "count": "30" 4 }</pre>
6	<pre>1 { 2 "step": "06.funds.received", 3 "count": "4", 4 "response": 200 5 }</pre>
7	<pre>1 { 2 "step": "07.notify.regulatory.board",</pre>

INGEST AND ALERT

Workflow Exercise 2

INGEST AND ALERT

Goal: Retrieve data from an external system and ingest it as a metric. If the value crosses a configured threshold, then generate an event/alert.

- Credential Vault
 - Access an API token from the Credential Vault using the Dynatrace SDK
- Metric Ingest
 - Retrieve data from an external system using HTTP GET
 - Ingest retrieved data point using the Dynatrace API with API token (not optimal)
 - Ingest retrieved data point using the Dynatrace SDK (optimal)
- Event Ingest
 - Query ingested metric data point using DQL
 - Evaluate metric value against a threshold
 - If the threshold is breached, generate an event using the Dynatrace SDK

Begin by creating a new Workflow in the Workflows App within your Dynatrace environment.

Workflows Executions

Workflow Essentials - Ingest and Alert

Modified

Save

Run

Executions

Trigger

Click to select a trigger



Select a trigger: choose
`On demand trigger`

Settings



Select a trigger

Triggers start the execution of a workflow. They can listen for events or be scheduled to run at fixed times or intervals.

Event



Event trigger

Run workflow based on a custom event filter.



Davis problem trigger

Run workflow based on problems detected by Davis.



Davis event trigger

Run workflow based on events detected by Davis.

Schedule



Fixed time trigger

Run workflow at a fixed time of day.



Time interval trigger

Run workflow at fixed time intervals.



Cron schedule trigger

Run workflow on a cron schedule.

On demand



On demand trigger

Use the Run button to trigger the workflow or use an API.

Workflow Essentials - Ingest and Alert

Modified

Save



Run

Executions



On demand trigger

Triggered manually or with an API



Click the '+' button to add a new action to the Workflow.



On demand trigger

Change trigger



Select Run to manually trigger the workflow or use an [API](#)

API script

```
1 // This is an example of a REST request using curl
2 curl \
3   -H 'Authorization: Bearer <YOUR_BEARER_TOKEN>' \
4   -X POST https://knp40672.sprint.apps.dynatracelabs.com/platform/automation/v
```

Note: Use your [OAuth2 client configuration](#) to get a valid bearer token for the request. It will require the `automation:workflows:run` scope.

Workflow Essentials - Ingest and Alert

Modified

Save

Run

Executions



On demand trigger

Triggered manually or with an API

New task

Select an action from the list



Choose action: choose
'Run JavaScript' action
type.

Choose action

Add a task to the workflow and select the action it performs

Search actions...

Workflows



Execute DQL Query
Executes DQL query



HTTP Request
Issue an HTTP request to any API



Run JavaScript
Build a custom task running js Code

Kubernetes Automations (Preview)



Apply configurations
Apply a configuration change to a resource or create new Kubernetes resources



Delete resources
Delete Kubernetes resources



Get logs
Get the logs for a container in a pod (limited to max. 1MB)



Get resource
Retrieve a Kubernetes resource



List resources
List Kubernetes resources



Restart deployment
Restart a Kubernetes deployment

Workflow Essentials - Ingest and Al...

Save

Run

Executions



Paste the following code snippet
into the action `Input`:

On demand trigger

Triggered manually or with an API

js_set_parameters

Set the parameters for this workflow

Set the value of the
missing variables



js_set_parameters

Set the parameters for this workflow

Change action



Input

Conditions

Options

Sample result

Source code

```
1 export default async function () {
2
3   let PARAMETERS = {
4
5     // EASYTRAVEL_URL - the URL to the easytravel rest API to search for journeys and receive JSON payload
6     EASYTRAVEL_URL: "http://aeb26f2507d934338baf67206067edef-612923892.us-east-1.elb.amazonaws.com/easytravel/rest/journeys/?n
7
8     // METRIC - the unique metric id to ingest a data point for the easytravel journey amount
9     METRIC: "custom.workflow_essentials.easytravel.journey_amount", // i.e. METRIC: "custom.workflow_essentials.easytravel.jou
10
11     // CREDENTIAL - the credential vault entity ID that holds the Dynatrace API token value with metrics.ingest scope
12     CREDENTIAL: "CREDENTIALS_VAULT-4D[REDACTED]", // i.e. CREDENTIAL: "CREDENTIALS_VAULT-123ABCF8F36FD"
13
14     // INGEST_URL - the URL of the dynatrace metric ingest API endpoint
15     INGEST_URL: "https://[REDACTED]sprint.dynatracelabs.com/api/v2/metrics/ingest", // i.e. INGEST_URL: "https://abc123.live.d
16
17     // THRESHOLD - the threshold amount to trigger an alert event
18     THRESHOLD: 500, // i.e. THRESHOLD: 500; trigger an alert event if the average journey amount is above 500.00
19
20   }
21
22   return PARAMETERS;
23 }
```

WorkflowsExecutions

Workflow Essentials - Ingest and Al... Save Run Executions

On demand trigger

Triggered manually or with an API

js_set_parameters

Set the parameters for this workflow

http_easytravel_search

Query external easytravel API for journey data

http_easytravel_search

Query external easytravel API for journey data

InputConditionsOptionsSample result

Method*GET

URL*js_set_parameters.EASYTRAVEL...
URL to which HTTP requests will be done

AuthenticationAuthenticate with credential vault Manage Credentials

Payload1

Headersacceptapplication/json

Error handling

http_easytravel_search

Query external easytravel API for journey data

InputConditionsOptionsSample result

Run this task ifjs_set_parameters

And custom condition was met

1{{ result("js_set_parameters")['EASYTRAVEL_URL'] is defined }}

ElseStop here

Choose action: choose 'HTTP Request' action type.

Configure the HTTP Request Input details

Set the task conditions

Click the '+' button to add a new action to the Workflow.

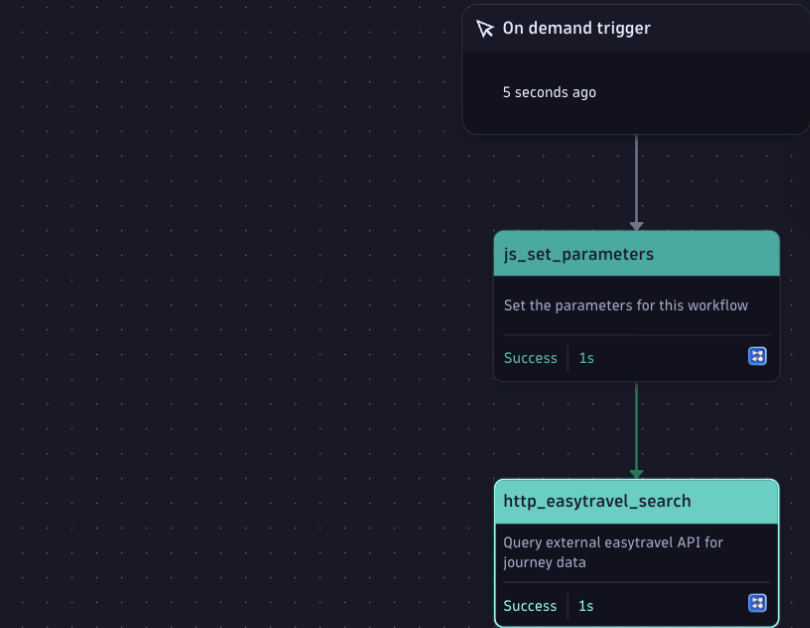
Workflow Essenti...

Success

Edit workflow

Cancel

Refresh



Run the workflow and validate the results

http_easytravel_search log

1	No log available.
---	-------------------

http_easytravel_search

Query external easytravel API for journey data

Cancel task

Detail Input Result

body

```
1 [{"id":2065936742,"name":"Robson - Dallas","fromDate":1472688000000,"toDate":14735520
```

json

```
1 [
2   {
3     "id": 2065936742,
4     "name": "Robson - Dallas",
5     "start": "Robson",
6     "amount": 1139.6000000000001,
7     "images": {
8       "nextPath": "images/6.jpg",
9       "currentPath": "images/4.jpg",
10      "previousPath": "images/14.jpg"
11    },
12    "tenant": "Personal Travel Inc.",
13    "toDate": 1473552000000,
14    "fromDate": 1472688000000
```

headers

```
1 {
2   "date": "Tue, 03 Sep 2024 16:08:32 GMT",
3   "server": "nginx/1.20.1",
4   "connection": "keep-alive",
5   "set-cookie": "dtCookie=v_4_srv_7_sn_B2AD7D849E0A9811EC4EABB23488471B_perc_100000_o1_0",
6   "content-type": "application/json",
7   "server-timing": "dtSInfo;desc=\"0\", dtRpid;desc=\"-261809590\"",
8   "traceresponse": "00-71afc50ed20e261326d0e124c98e90ea-f969dc6338f9a13c-01",
9   "x-dt-tracestate": "4ec8866d-ba93df2a@dt",
10  "transfer-encoding": "chunked",
11  "x-oneagent-js-injection": "true",
12  "access-control-allow-origin": "*",
13  "access-control-allow-headers": "origin, content-type, accept, authorization",
14  "access-control-allow-methods": "GET, POST, PUT, DELETE, OPTIONS, HEAD",
```

status_code

200

Workflow Essentials - Ingest and Alert

Save

Run

Executions

⋮

Choose action: choose `Run JavaScript` action type.

Workflow trigger

Triggered manually or with an API

js_set_parameters

Set the parameters for this workflow

http_easytravel_search

Query external easytravel API for journey data

js_get_credential

Access API token from the credential vault

Click the `+` button to add a new action to the Workflow.



js_get_credential

Access API token from the credential vault

Input

Conditions

Options

Sample result

Paste the following code snippet into the action `Input`:

Source code

```
1 import { execution } from '@dynatrace-sdk/automation-utils';
2 import { credentialVaultClient } from '@dynatrace-sdk/client-classic-environment-v2';
3
4 const PARAMETERS_TASK = 'js_set_parameters';
5
6 export default async function ({ execution_id }) {
7
8   // get parameters from previous tasks
9   // execution
10  const ex = await execution(execution_id);
11  // parameters
12  const parameters = await ex.result(PARAMETERS_TASK);
13  // CREDENTIAL
14  const CREDENTIAL = parameters['CREDENTIAL'];
15
16  // get the credentials from the credential vault using the SDK
17  // https://developer.dynatrace.com/develop/sdks/client-classic-environment-v2/#getcredentialsdetails
18  const data = (await credentialVaultClient.getCredentialsDetails({id: CREDENTIAL}));
19  const token = data['token'];
```



js_get_credential

Access API token from the credential vault

Input

Conditions

Options

Sample result

Run this task if

http_easytravel_search

Set the task conditions

success



And custom condition was met

```
1 {{ result("http_easytravel_search")["status_code"] == 200 and result("http_easytravel_search")["json"] | length > 0 }}
```

Else

Stop here

Workflow Essentials - Ingest and Alert

Save

Run

Executions

On demand trigger

Triggered manually or with an API

js_set_parameters

Set the parameters for this workflow

http_easytravel_search

Query external easytravel API for journey data

js_get_credential

Access API token from the credential vault

http_ingest_metric

Ingest a metric data point using the metric API

Click the '+' button to add a new action to the Workflow.



http_ingest_metric

Ingest a metric data point using the metric API

Input Conditions Options Sample result

Method*

POST

URL*

js_set_parameters.INGEST_URL

URL to which HTTP requests will be done

Authentication

☒Authenticate with credential vault [Manage Credentials](#)

Payload

1

js_set_parameters.METRIC , id= http_easytravel_search.json[0].id , destination= http_easytravel_search.json[0]....

Headers

Content-Type

text/plain

Authorization

Api-Token js_get_credential.token

Enter key...

Enter value...

Choose action: choose 'HTTP Request' action type.

Configure the HTTP Request Input details



http_ingest_metric

Ingest a metric data point using the metric API

Input Conditions Options Sample result

Run this task if

js_get_credential

success

And custom condition was met

1

{{ result("js_get_credential")['token'] is defined }}

Else

Stop here

Set the task conditions

Workflow Essential...

Success

Edit workflow

Cancel

Refresh



js_set_parameters

Set the parameters for this workflow

Success 1s



http_easytravel_search

Query external easytravel API for journey data

Success 1s



js_get_credential

Access API token from the credential vault

Success 1s



http_ingest_metric

Ingest a metric data point using the metric API

Success 1s



Run the workflow and
validate the results



http_ingest_metric

Ingest a metric data point using the metric API

Cancel task

Detail

Input

Result

body

1

```
{"linesOk":1,"linesInvalid":0,"error":null,"warnings":null}
```



json

1

2

3

4

5

6

```
{
  "error": null,
  "linesOk": 1,
  "warnings": null,
  "linesInvalid": 0
}
```



headers

1

2

3

4

5

6

7

8

9

10

11

12

13

14

```
{
  "date": "Tue, 03 Sep 2024 18:40:42 GMT",
  "vary": "Accept-Encoding",
  "pragma": "no-cache",
  "server": "ruxit gateway",
  "set-cookie": "dtCookie=v_4_srv_2_sn_0939EAD4D806C2888C2FCDED48FFB924_perc_100000_o",
  "content-type": "application/json;charset=utf-8",
  "x-robots-tag": "noindex",
  "cache-control": "no-store, no-cache",
  "server-timing": "dtRpid;desc=\"1616655408\", dtSInfo;desc=\"0\"",
  "traceresponse": "00-e53bc9b2ec07c87da1b869431fc211c3-bdbc60b415cfd91e-01",
  "x-dt-tracestate": "f148e083-4b338275@dt",
  "x-oneagent-js-injection": "true",
  "strict-transport-security": "max-age=31536000;includeSubDomains"
```

status_code

202

Workflow Essentials - Ingest and Alert

Save

Run

Executions

Choose action: choose `Run JavaScript` action type.

On demand trigger

Triggered manually

js_set_parameters

Set the parameters for this workflow

http_easytravel_search

Query external easytravel API for journey data

</>

js_get_credential

Access API token from the credential vault

</>

http_ingest_metric

Ingest a metric data point using the metric API

</>

Click the `+` button to add a new action to the Workflow.

js_ingest_metric_sdk

Ingest a metric data point using the SDK

</>



js_ingest_metric_sdk

Ingest a metric data point using the SDK

Input

Conditions

Options

Sample result

Paste the following code snippet into the action `Input`:

Source code

```
1 import { execution } from '@dynatrace-sdk/automation-utils';
2 import { metricsClient } from '@dynatrace-sdk/client-classic-environment-v2';
3
4 const PARAMETERS_TASK = 'js_set_parameters';
5 const SEARCH_TASK = 'http_easytravel_search';
6
7 export default async function ({ execution_id }) {
8
9   // get parameters from previous tasks
10  // execution
11  const ex = await execution(execution_id);
12  // parameters
13  const parameters = await ex.result(PARAMETERS_TASK);
14  const METRIC = parameters['METRIC'];
15  // previous results
16  const search_results = await ex.result(SEARCH_TASK);
17  const id = search_results['json'][0]['id'];
18  const amount = search_results['json'][0]['amount'];
19  const destination = search_results['json'][0]['destination'];
20 }
```



js_ingest_metric_sdk

Ingest a metric data point using the SDK

Input

Conditions

Options

Sample result

Run this task if

http_easytravel_search

Set the task conditions

success



And custom condition was met

```
1 {{ result("http_easytravel_search")["status_code"] == 200 and result("http_easytravel_search")["json"] | length > 0 }}
```

Else

Stop here

Workflow Essentials - Inges...

Success

Edit workflow

Cancel

Refresh



js_set_parameters

Set the parameters for this workflow

Success 1s



http_easytravel_search

Query external easytravel API for journey data

Success 1s



js_get_credential

Access API token from the credential vault

Success 1s



http_ingest_metric

Ingest a metric data point using the metric API

Success 1s



js_ingest_metric_sdk

Ingest a metric data point using the SDK

Success 1s

Run the workflow and
validate the results



js_ingest_metric_sdk

Ingest a metric data point using the SDK

Cancel task

Detail Input Result

data

```
1 {
2   "linesOk": 1,
3   "linesInvalid": 0
4 }
```



js_ingest_metric_sdk log

```
1 custom.workflow_essentials.easytravel.journey_amount,id=2065936742,destination=Dallas
2 1139.6000000000001
```

Workflow Essentials - Ingest and Alert

Save Run Executions

js_set_param

Set the param

Choose action: choose 'Run JavaScript' action type.

http_easytravel_search

Query external easytravel API for journey data

</> +

js_get_credential

Access API token from the credential vault

</> +

http_ingest_metric

Ingest a metric data point using the metric API

</> +

Click the '+' button to add a new action to the Workflow.

js_wait_30_seconds

Wait 30 seconds for data to be processed

+



js_wait_30_seconds

Wait 30 seconds for data to be processed

Change action

Input Conditions Options Sample result

Source code

```
1 export default async function () {
2
3   const sleep = async (waitTime: number) =>
4     new Promise(resolve =>
5       setTimeout(resolve, waitTime));
6
7   const waitTime = async () => {
8     // 30,000ms = 30 seconds to sleep
9     await sleep(30000);
10    console.log("Finished Waiting");
11  }
12
13  console.log("Waiting 30 Seconds");
14  waitTime();
15
16 }
```

Paste the following code snippet into the action 'Input':



js_wait_30_seconds

Wait 30 seconds for data to be processed

Change action

Input Conditions Options Sample result

Run this task if

js_ingest_metric_sdk

success

http_ingest_metric

success

Set the task conditions

And custom condition was met

1

Else

Stop here

This will be the task/action that pauses workflow execution for 30 seconds with a timed wait using JS code.

Additionally, you can configure any task to wait up to 60 seconds in the 'Options' tab.

Source code

```
1  export default async function () {  
2  
3      const sleep = async (waitTime: number) =>  
4          new Promise(resolve =>  
5              setTimeout(resolve, waitTime));  
6  
7      const waitTime = async () => {  
8          // 30,000ms = 30 seconds to sleep  
9          await sleep(30000);  
10         console.log("Finished Waiting");  
11     }  
12  
13     console.log("Waiting 30 Seconds");  
14     waitTime();  
15  
16 }
```



js_wait_30_seconds

Wait 30 seconds for data to be processed

↩ Change action



Input

Conditions

Options

Sample result



Wait before ||

Controls how long a task stays in waiting state before being run. Default set to 0 second.

Wait (seconds)

30



Workflow Essentials - Ingest and Alert



Executions



Choose action: choose 'Run JavaScript' action type.

js_get_credential

Access API token from the credential vault

</>

http_ingest_metric

Ingest a metric data point using the metric API

</>

js_wait_30_seconds

Wait 30 seconds for data to be processed

js_dql_query_alert

Query metric data point and generate an alert via SDK

js_ingest_metric_sdk

Ingest a metric data point using the SDK

</>



js_dql_query_alert

Query metric data point and generate an alert via SDK

Input

Conditions

Options

Sample result

Source code

```
1 import { execution } from '@dynatrace-sdk/automation-utils';
2 import { queryExecutionClient } from '@dynatrace-sdk/client-query';
3 import { eventsClient, EventIngestEventType } from '@dynatrace-sdk/client-classic-environment-v2';
4
5 const PARAMETERS_TASK = 'js_set_parameters';
6 const SEARCH_TASK = 'http_easytravel_search';
7
8 export default async function ({ execution_id }) {
9
10   // get parameters from previous tasks
11   // execution
12   const ex = await execution(execution_id);
13   // parameters
14   const parameters = await ex.result(PARAMETERS_TASK);
15   const METRIC = parameters['METRIC'];
16   const THRESHOLD = parameters['THRESHOLD'];
17   // previous results
18   const search_results = await ex.result(SEARCH_TASK);
19   const destination = search_results['json'][0]['destination'];
20 }
```

Paste the following code snippet into the action 'Input':



js_dql_query_alert

Query metric data point and generate an alert via SDK

Input

Conditions

Options

Sample result

Run this task if

js_wait_30_seconds

success



And custom condition was met

1

Else

Stop here



Click the '+' button to add a new action to the Workflow.

Set the task conditions

Workflow Essentials - ...

Success

Edit workflow

Cancel

Refresh

Query external easytravel API for journey data

Success

1s



js_get_credential

Access API token from the credential vault

Success

1s



http_ingest_metric

Ingest a metric data point using the metric API

Success

1s



js_ingest_metric_sdk

Ingest a metric data point using the SDK

Success

1s



js_wait_30_seconds

Wait 30 seconds for data to be processed

Success

30s



js_dql_query_alert

Query metric data point and generate an alert via SDK

Success

1s



Run the workflow and validate the results

js_dql_query_alert log

1

No log available.



js_dql_query_alert

Query metric data point and generate an alert via SDK

Cancel task

Detail

Input

Result

amount

1139.6000000000001

threshold

500

correlationId

1

ad32a52cba544d6d





TPCSprint: EasyTravel Journey Amount [1139.6000000000001]

> Custom alert P-24091 detected at 13:53 (open for 51 seconds).

[Share feedback](#)

[Close problem](#)



Affected applications

–



Affected services

–



Affected infrastructure

–



Affected SLOs

–



DAVIS®
analyzed dependencies
across 1 monitored entities

Impact



TPCSprint
Environment

EasyTravel Journey Amount [1139.6000000000001]

EasyTravel Journey Amount [1139.6000000000001]

[\(Show less\)](#)

amount

1139.6000000000001

threshold

500

Open the Problems (or Problems (Classic))
app and view the open problem alert.

Comments



Comment on this problem

[Add comment](#)

No comments posted

THIRD PARTY INTEGRATION

Workflow Exercise 3

THIRD PARTY INTEGRATION

Goal: When a Davis problem event is triggered, send a problem summary as a Slack message to a chosen Slack channel.

- Event Trigger
 - Trigger Workflow when a Davis problem event is detected
 - Events are filtered on the type of event, based on the previous exercise
- Slack for Workflows
 - Third party integration is typically accomplished with Workflow Apps
 - The Slack for Workflows App provides an easy-to-use mechanism for sending messages to a Slack environment containing the data and context from Dynatrace
 - Leverage Jinja expressions to access Workflow action results and implement logic

Begin by creating a new Workflow in the Workflows App within your Dynatrace environment.

Workflows Executions

Workflow Essentials - Slack Integration

Save Run Executions

Davis problem trigger

On: active problem, or closed; Severity: custom

+

Select a trigger: choose `Davis problem trigger`

Filter the events that will trigger this workflow.

Davis problem trigger

Run workflow based on problems detected by Davis.

Event state*

active or closed

Event category*

Custom

Learn more about [event categories](#).

Affected entities*

include entities with all defined tags below

Tags

Enter key... Enter value...

Additional custom filter query

Filter query

1

matchesPhrase(event.name, "EasyTravel Journey Amount")

Query past events

Workflow Essentials - Slack Integration

Modified

Save

Run

Executions ▾



Davis problem trigger



On: active problem, or closed; Severity: custom

Click the '+' button to add a new action to the Workflow.

task

an action from the list



Choose action

Add a task to the workflow and select the action it performs

Search actions...

Workflows



Execute DQL Query

Executes DQL query



HTTP Request



Run JavaScript

Build a custom task running js Code

Choose action: choose 'Run JavaScript' action type.

Kubernetes Automations (Preview)



Apply configurations

Apply a configuration change to a resource or create new Kubernetes resources



Delete resources

Delete Kubernetes resources



Get logs

Get the logs for a container in a pod (limited to max. 1MB)



Get resource

Retrieve a Kubernetes resource



List resources

List Kubernetes resources



Restart deployment

Restart a Kubernetes deployment

Ownership



(Deprecated) Get contact details

Extracts a list of contact details from teams that are returned by the "get_owners" workflow action. Additionally, the contact details can be filtered by the contact type.



Get owners

Retrieves owners from Entities and Team identifiers.



Import teams

Import teams into Dynatrace.

Site Reliability Guardian



Site Reliability Guardian

Automation action to start a Site Reliability Guardian validation

Slack for Workflows



Send message

Send a message to a Slack workspace

Workflow Essentials - Slack Integration

Save

Run

Executions



Paste the following code snippet
into the action `Input`:

Davis problem trigger

On: active problem, or closed; Severity:
custom

js_set_parameters

Set the parameters for this workflow



js_set_parameters

Set the parameters for this workflow

Change action



Input

Conditions

Options

Sample result

Source code

```
1 export default async function () {  
2  
3 let PARAMETERS = {  
4  
5 // SLACK_CHANNEL - the Slack channel name (without the #) that the Slack bot has been added to  
6 SLACK_CHANNEL: "tpc-workflow-essentials", // i.e. SLACK_CHANNEL: "ace-workflow-essentials",  
7  
8 }  
9  
10 return PARAMETERS;  
11 }
```

Set the value of the
missing variables

Workflow Essentials - Slack Integrat...

Modifi...

Save

Run

Executions



Davis problem trigger



On: active problem, or closed; Severity:
custom

js_set_parameters

Set the parameters for this workflow



New task

Select an action from the list



Choose action: choose
'Send message' action
type.

Choose action

Add a task to the workflow and select the action it performs

Search actions...



Execute DQL Query

Executes DQL query



HTTP Request

Issue an HTTP request to any API



Run JavaScript

Build a custom task running Js Code

Kubernetes Automation (Preview)



Apply configurations

Apply a configuration change to a resource or create new Kubernetes resources



Delete resources

Delete Kubernetes resources



Get logs

Get the logs for a container in a pod (limited to max. 1MB)



Get resource

Retrieve a Kubernetes resource



List resources

List Kubernetes resources



Restart deployment

Restart a Kubernetes deployment



Wait

Waits for a condition on a specified Kubernetes resource

Ownership



(Deprecated) Get contact details

Extracts a list of contact details from teams that are returned by the "get_owners" workflow action. Additionally, the contact details can be filtered by the contact type.



Get owners

Retrieves owners from Entities and Team identifiers.



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Site Reliability Guardian

Automation action to start a Site Reliability Guardian validation

Slack for Workflows



Send message

Send a message to a Slack workspace

Workflow Essentials - Slack Integrat...

Modifi...

Save

Run

Executions ▾

⋮

Configure the Send message action 'Input':

 Davis problem trigger 

On: active problem, or closed; Severity: custom

js_set_parameters

Set the parameters for this workflow



slack_send_message

Send problem summary in Slack



slack_send_message

Send problem summary in Slack

Change action

⋮

Input

Conditions

Options

Sample result

Connection*

workflow_essentials_demo ▾

[Create a new connection](#)

Channel*

`{{ result("js_set_parameters")['SLACK_CHANNEL'] }}` ✕

Message*

1

Note: You can use [Slack Markdown](#) within your message, or paste the result of the [Block Kit Builder](#) .

Send Test Message ▶

Reactions

Please select one or multiple reactions ▾

Message options

☒ Append to thread

Send the message as a reply to a previous message, which has been sent from within the same workflow execution.

Attachments

+ Attachment

Workflow Essentials - Slack Integration

Save

Run

Executions



Davis problem trigger

On: active problem, or closed; Severity:
custom

js_set_parameters

Set the parameters for this workflow



slack_send_message

Send problem summary in Slack

Input Conditions Options Sample result

Run this task if

js_set_parameters

success



And custom condition was met

1

{{ result("js_set_parameters")['SLACK_CHANNEL'] is defined }}

Else

Stop here



slack_send_message

Send problem summary in Slack

Change action



Input Conditions Options Sample result

Connection*

workflow_essentials_demo

[Create a new connection](#)

Channel*

{{ result("js_set_parameters")['SLACK_CHANNEL'] }}



Message*

```
1 {
2   "blocks": [
3     {
4       "type": "section",
5       "block_id": "sectionHeader",
6       "text": {
7         "type": "mrkdwn",
8         "text": "Dynatrace :dt: Problem Detection \n Workflow Execution: [ execution.id ]"
9       }
10    }
11  ]
12 }
```

Change action



Set the task conditions

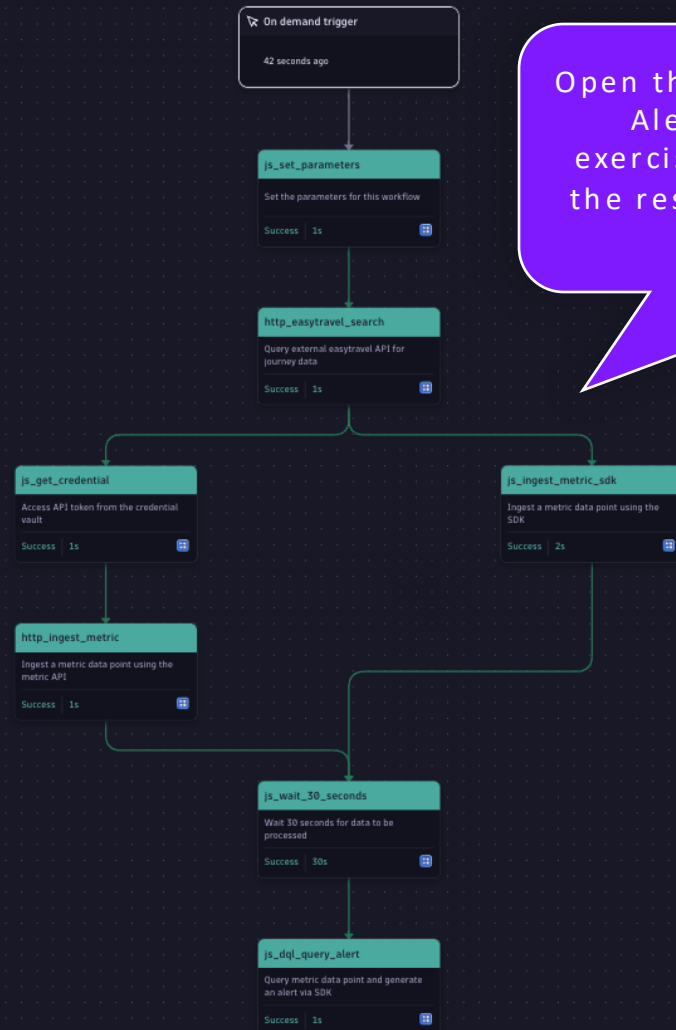
Workflow Essentials - Ingest and Al...

Success

Edit workflow

✕ Cancel

↻ Refresh



Open the Workflow Essentials - Ingest and Alert workflow from the previous exercise. Run the workflow and validate the results. A new problem event should have been created.

Workflow Essentials - Slack Integr...

Success

Edit workflow

Cancel

Refresh

Workflow Essentials - Slack Integration

Execution ID: 67336348-c787-47be-8801-7e36682293e3

Detail Input Params

```
1 {
2   "event.id": "-5044649459813518699_1725463178682V2",
3   "timestamp": "2024-09-04T15:19:41.927000000Z",
4   "display_id": "P-24092",
5   "event.kind": "DAVIS_PROBLEM",
6   "event.name": "EasyTravel Journey Amount [ 1139.600000000",
7   "event.start": "2024-09-04T15:19:38.682000000Z",
8   "event.status": "ACTIVE",
9   "event.category": "CUSTOM_ALERT",
10  "dt.system.bucket": "default_davis_events",
11  "event.description": "# EasyTravel Journey Amount [ 113",
12  "dt.davis.event_ids": [
13    "-5044649459813518699_1725463178682"
14  ],
15 }
```

Shortly after running the other workflow, the Workflow Essentials - Slack Integration workflow should have triggered as a result of the problem event. Check the execution and validate the results.

Event trigger

1 minute ago

js_set_parameters

Set the parameters for this workflow

Success 1s

slack_send_message

Send problem summary in Slack

Success 1s

Dynatrace  Problem Detection

Workflow Execution: [67336348-c787-47be-8801-7e36682293e3]

Problem State:  OPEN 

Problem Number: [P-24092]

Problem ID: [-5044649459813518699_1725463178682V2]

[EasyTravel Journey Amount [1139.6000000000001]]

Open the Slack channel and validate that the message contains the problem summary.

View Problem

View Workflow Execution

Search all apps, extensions, and technologies

Here you'll find hundreds of technologies to tame digital complexity and get more value from data, from application and infrastructure monitoring to digital experience and application security.

Third party integration is typically accomplished with Workflow Apps. Available Apps are found in the Hub.

Hub home

Get data into Dynatrace

Build your own app

All (17)

Log Management and Analytics

Infrastructure Observability

Automations

Application Observability

Application Security

Digital Experience

Business Analytics

Search apps, extensions, technologies

Type: Any

Actions



Azure for Workflows (Preview)

Installed

App 1 Actions

Fetch groups from Microsoft Entra ID (formerly Azure Active Directory).



Davis® for Workflows

Installed

App 1 Actions

Implement custom AI analysis in Grail data lakehouse using Davis® for Workflows.



DevSecOps Lifecycle Coverage with Snyk

App 1 Actions

Connect Snyk container scans and Dynatrace runtime insights for strong security.



Email for Workflows (Preview)

Installed

App 1 Actions

Send out-of-the-box targeted Emails with workflows.



GitHub for Workflows (Preview)

App 7 Actions

Amend files in GitHub repositories for GitOps-based remediation



GitLab for Workflows

Installed

App 14 Actions

Amend files in GitLab repositories, trigger pipelines, and manage issues.



Jira for Workflows

Installed

App 5 Actions

Create, query, comment, transition, and resolve Jira tickets within workflows.



Microsoft 365 for Workflows (Preview)

Installed

App 1 Actions

Send targeted emails from workflows using your Microsoft 365 account.



Microsoft Teams for Workflows (Preview)

Installed

App 1 Actions

Automate Microsoft Teams messaging for problems, security incidents, and more.



Ownership

Installed

App 3 Actions

Enables the import of and access to ownership information of entities.



PagerDuty for Workflows

Installed

App 6 Actions

Create incidents in PagerDuty and list all of the on-calls



Red Hat Ansible for Workflows (Preview)

Installed

App 4 Actions

Integrate with Red Hat Event Driven Ansible, run job templates or retry a job



ServiceNow for Workflows

Installed

App 8 Actions

Import groups and create, search, comment, and resolve incidents in ServiceNow.



Site Reliability Guardian

Installed

App 1 Actions

SRG: Automated change impact analysis for your deployment and release processes



Slack for Workflows

Installed

App 1 Actions

Automate Slack messaging for security incidents, attacks, remediation, and more.



Synthetic for Workflows

Installed

App 1 Actions

Execute Dynatrace synthetic monitors within Workflows.



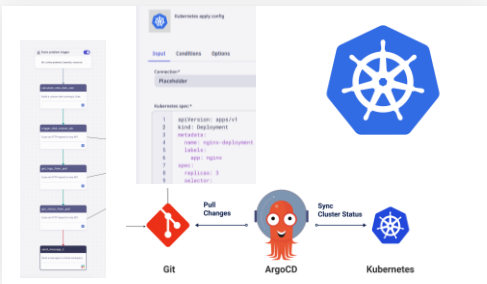
Workflows

Installed

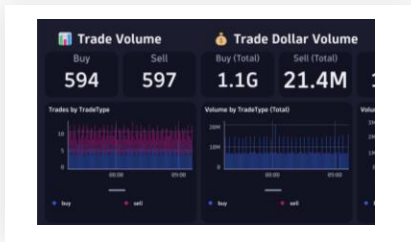
App 3 Actions

Automate tasks in your IT landscape, remediate problems, and visualize processes

Automations Roadmap Highlights



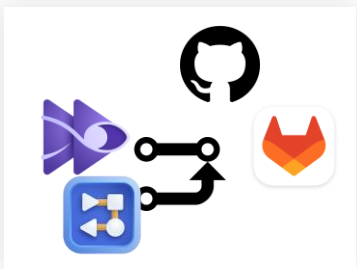
Cloud-Native Automation
Automate using the K8s API



Config-as-Code
for new **Dashboards**



Azure Workflow Actions
Hyperscaler workflow actions



GitOps Automation
Changing configurations
with Git Pull Requests

SOON

NEXT

LATER

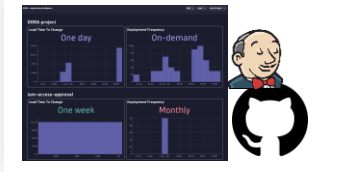
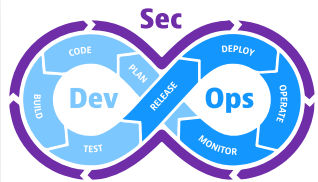


AWS Workflow Actions
Hyperscaler workflow
actions for automated
resizing and remediation

SLOs on all data types
Leveraging the full power of
Grail for SLO definition

Service-level objectives		
Name	Status	Target
Cluster memory efficiency	35.76%	80%
Memory requests efficiency	67.89%	80%

**SDLC observability
& analytics foundation**
Dedicated Grail software development
lifecycle (SDLC) events



**Out-of-the-box dashboards for
engineering KPIs (DORA)** with
events from GitHub, ArgoCD,
Jenkins



Workflow Action EcoSystem

Delivered		
	Slack	2023-04
	Jira	2023-06
	PagerDuty	2023-08
	MS Teams	2023-10
	RH Ansible (OOS)	2023-10
	Microsoft365	2023-11
	ServiceNow	2023-12
	Email	2024-02
	RH Ansible (AAP / EDA)	2024-04
	GitLab (Issues)	2024-05

Planned		
	GitLab (Git)	FY25Q2
	GitHub	FY25Q3
	AWS	FY25Q3
	K8s	FY25Q4
	Jenkins	NEXT
	Azure	NEXT
	Opsgenie	NEXT
	User Approval	LATER
	Dynatrace SDK Actions	LATER
	Formatted E-Mail	LATER





Notebooks

+ Notebook

Upload



Workflow Essentials - Code Snippets



Share



History



Rerun sections

The Workflow Essentials – Code Snippets Notebook is a great reference guide to add to your Dynatrace environment

Workflow Parameters: Set Parameter Task

The first task of a workflow should be a parameter task that defines the parameters/variables used by the remainder of tasks in the workflow. Environment and deployment unique settings, like URLs, entities, etc. should be set as parameters in this task so that no other changes need to be made within the workflow when it is copied/cloned/moved for other uses.

The parameters are specified in a JSON object variable and then returned during task execution to be used by remaining tasks. Include a description/purpose for the parameter and provide an example value.

```
1 export default async function () {
2
3   let PARAMETERS = {
4     // PARAMETER - description of the parameter
5     PARAMETER: "VALUE", // i.e. PARAMETER: "VALUE" - example value of the parameter
6   }
7
8   return PARAMETERS;
9 }
```

Workflow Parameters: Using Parameters from Parameter Task

Tasks executed after the parameter task can access the parameters and values from the task result.

Within the JS code task, the `execution` class must be used from the `dynatrace-sdk/automation-utils` package. Access the parameter from the task result and assign the value to a new variable.

Within other task types (non JS code), the parameters can be accessed via Jinja expressions.

```
{{ result("js_set_parameters")['PARAMETER NAME'] }}
```

WORKFLOW BEST PRACTICES

- Make use of a parameters task as the first task in the workflow
 - Set every possible variable in the parameters task and use them in subsequent tasks
 - Don't send someone a workflow where they have to modify anything else!
- Use the HTTP Request action type when it fits your needs, but leverage the Fetch API (JavaScript) when it doesn't
 - The Fetch API provides additional flexibility and can be used multiple times in a single action
- Make use of task conditions to avoid unexpected errors
- Use the Dynatrace SDK to manage settings, configurations, and data
 - Don't use the Dynatrace API + Auth Token, waste of time and effort!
- Leverage task loops to execute logic on data/result sets
 - Individual loop executions can succeed while others fail, allowing you to handle errors
- Developer Docs (developer.dynatrace.com) are extremely helpful!

ASSESSMENT

SE Action Items

ASSESSMENT ACTION ITEM #1

Create a Workflow using materials provided in this lab

- Demonstrate that you understand the Workflow Essentials from this lab
- Use the BizEvent data that's being generated
 - Try using Notebooks and DQL to explore the data and find a use case
- Integrate with an external API of your choosing
 - Bring data in or send data out; your choice
 - Use Fetch API or the HTTP Request action type
- Make use of the Dynatrace SDK
 - Ingest data, manage configurations, perform an audit; your choice

ASSESSMENT ACTION ITEM #2

Create a Workflow to solve for a customer use case

- Pick a customer to help solve for a use case
 - SaaS customer is preferred
 - Zero to low adoption of Workflows is preferred
- Identify a challenge that's not being solved for, or is solved for manually
 - Integrating data and tool sets in their ecosystem
 - Acting on data and answers provided by Dynatrace
 - Product gaps requiring manual workarounds
- Build a Workflow to solve for their use case, conceptually
 - Doesn't have to be a production ready, fully scaled and tested solution
 - Build it in their environment or in your own sandbox (use your best judgement!)
 - Demonstrate that you understand the Workflow Essentials from this lab

ASSESSMENT ACTIONS

During the training, 3 Workflows will be created by the attendees.
After the training, 2 Workflows will need to be created by the attendees.

Attendees will have until Friday, November 15th to complete the 5 Workflows.

The 5 Workflows should be exported from the Dynatrace environment, zipped into an archive, named 'LASTNAME_FIRSTNAME_WORKFLOW_ESSENTIALS.zip', and sent to their Regional Director for collection/review.

DTU environments will be deleted 90 minutes after each session. Assessment action items will need to be completed in a separate Dynatrace tenant, such as their personal sprint tenant.

The background features a deep blue field with subtle, concentric wavy lines. In the top-left corner, there is a bright purple wavy line, and in the bottom-right corner, there is a bright blue wavy line.

THANK YOU